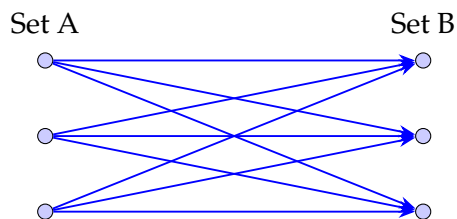


Equivalence Relations, Equivalence Classes, Partitions, and Quotient Sets

October 22, 2024

Cartesian Products / Set Products



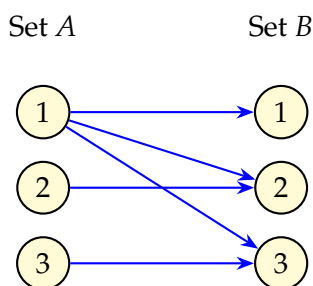
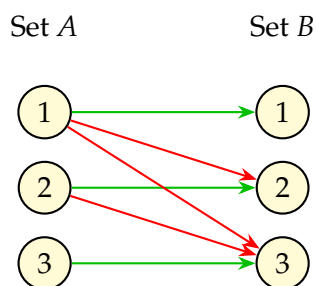
Example. If $A = \{1, 2\}$ and $B = \{x, y\}$, then:

$$A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$$

Definition. Let A and B are sets.

$$A \times B := \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Relations



Green arrows: Equality (=)

Red arrows: Less than (<)

Blue arrows: Divisibility (|)

Definition. Let A and B are sets. A **relation** $\mathcal{R}$ from A to B is a *subset* of the Cartesian Product $A \times B$:

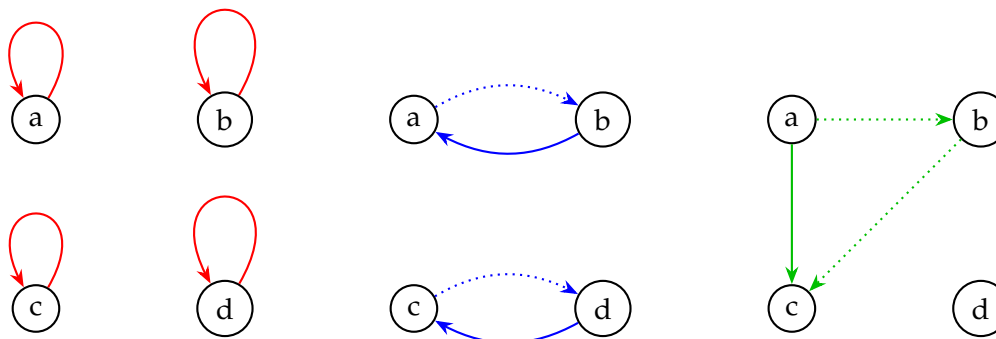
$$\mathcal{R} \subseteq A \times B$$

Notation.

$$(a, b) \in \mathcal{R} \subseteq A \times B \iff a \mathcal{R} b.$$

For example, $a = b \iff (a, b) \in =$

Equivalence Relations



Red arrows: Reflexivity

(each element is related to itself).

Blue arrows: Symmetry

(if a is related to b , then b is related to a).

Green arrows: Transitivity

(if a is related to b and b is related to c , then a is related to c).

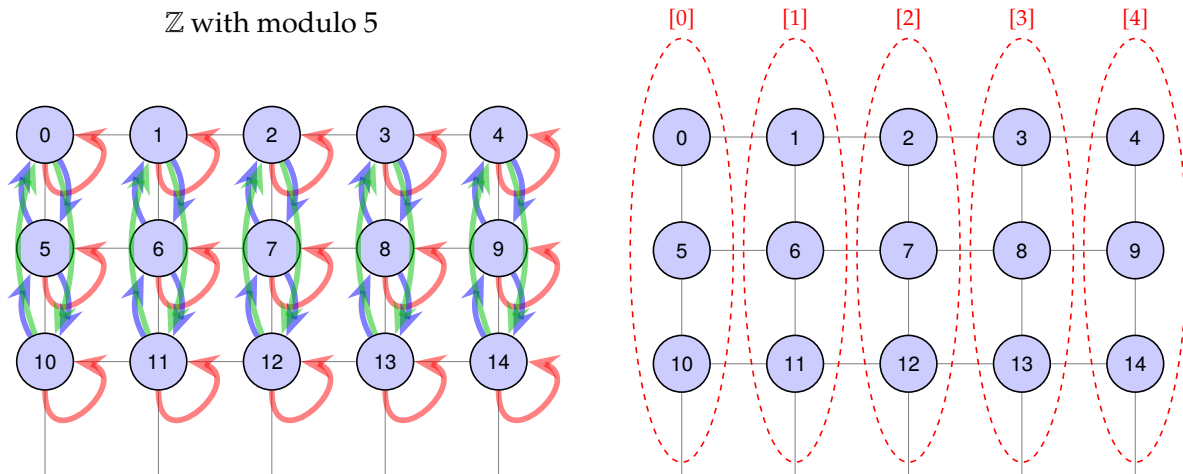
Example. Equality on $\mathbb{Z}$:

- $a \in \mathbb{Z} \implies a = a$;
- $a = b \implies b = a$;
- $a = b$ and $b = c \implies a = c$.

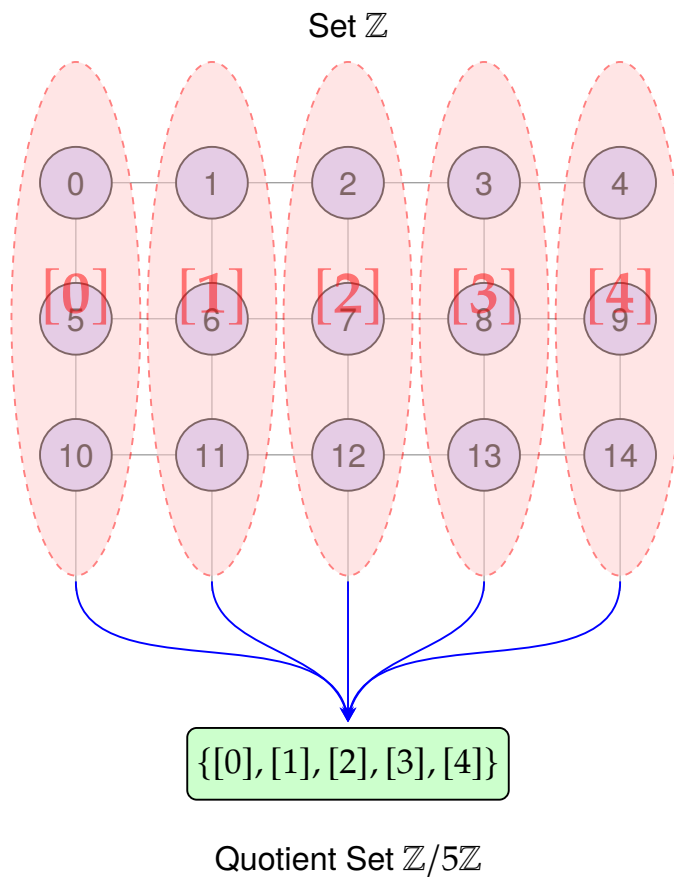
Definition. Given a set A , a relation $\mathcal{R} \subseteq A \times A$ is called an **equivalence relation** if it satisfies following properties:

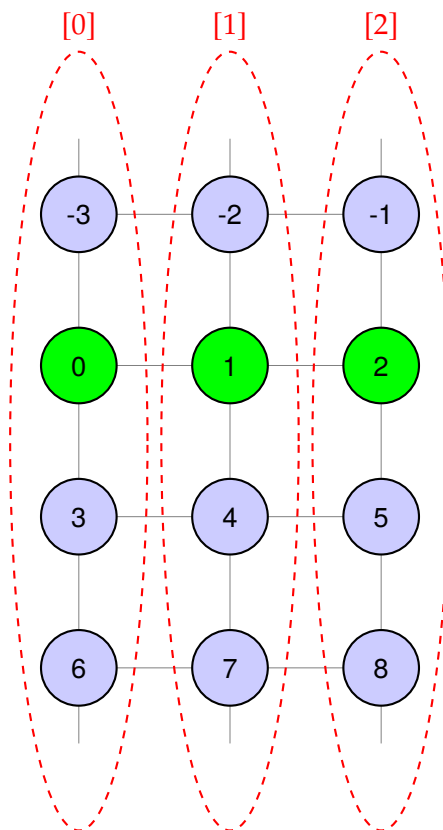
- (Reflexivity) $a \in A \implies (a, a) \in \mathcal{R}$;
- (Symmetry) $(a, b) \in \mathcal{R} \implies (b, a) \in \mathcal{R}$;
- (Transitivity) $(a, b) \in \mathcal{R} \wedge (b, c) \in \mathcal{R} \implies (a, c) \in \mathcal{R}$.

Equivalence Classes

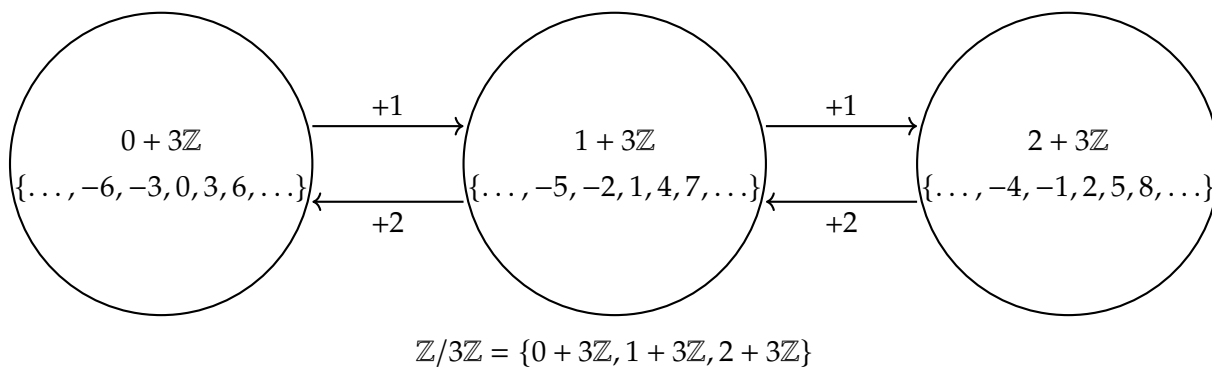
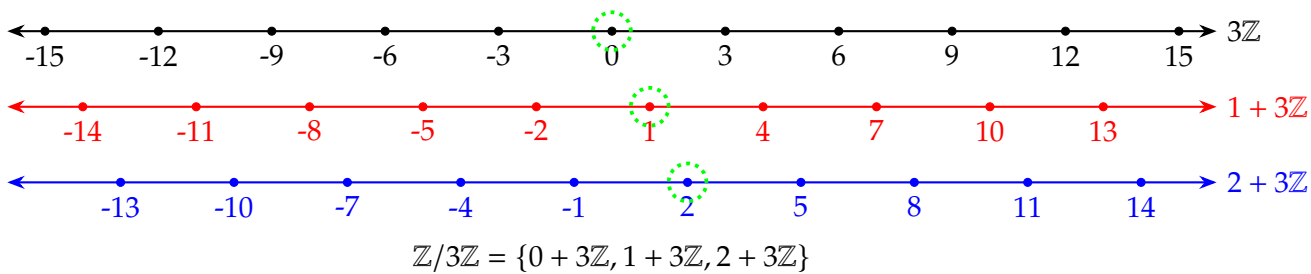


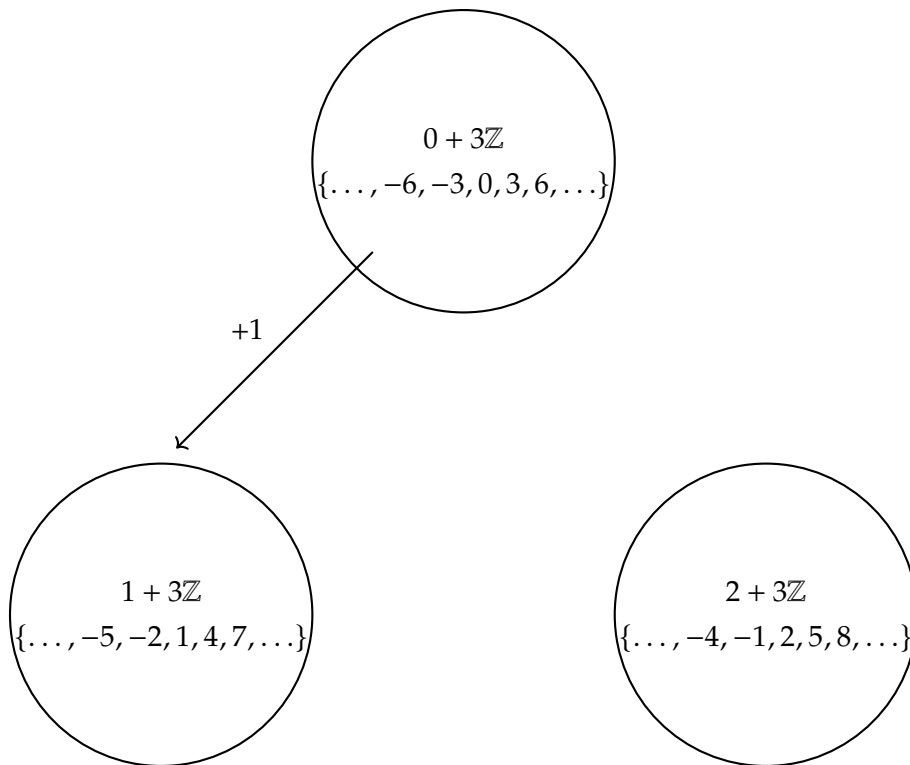
Quotient Sets





$$\mathbb{Z}/\sim = \{[0], [1], [2]\} \text{ with } a \sim b \stackrel{\text{def}}{\iff} a \equiv b \pmod{3}$$



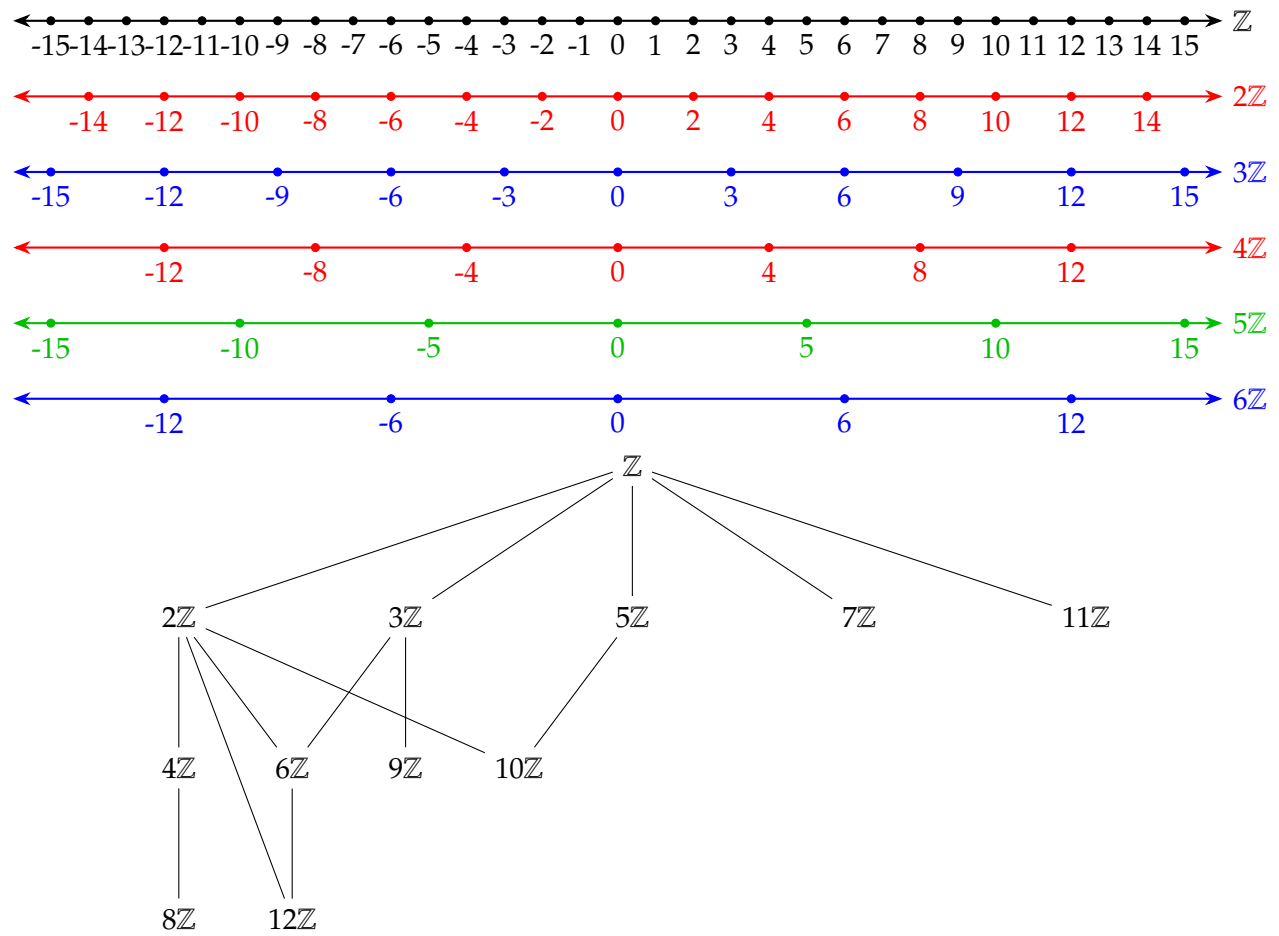


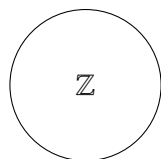
Groups

Subgroups

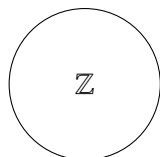
Cyclic Subgroups

Normal Subgroups

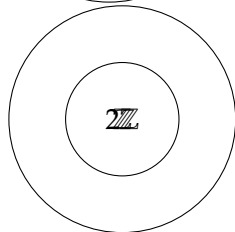
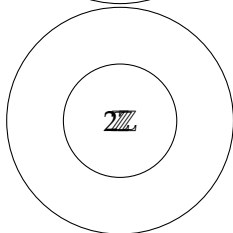
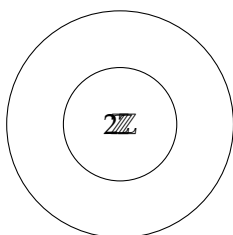




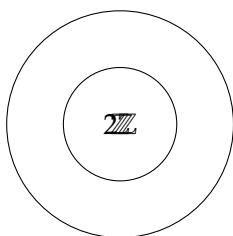
Operation: +
Identity: 0
Inverses: $\forall a \in \mathbb{Z}, -a \in \mathbb{Z}$



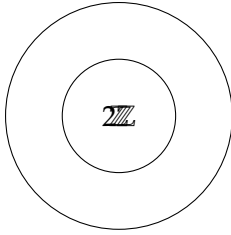
Addition: + Multiplication: $\cdot$
Additive Identity: 0 Multiplicative Identity: 1
Additive Inverses: $\forall a \in \mathbb{Z}, -a \in \mathbb{Z}$



Normal Subgroup

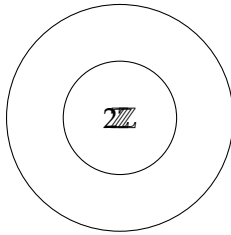


Ideal



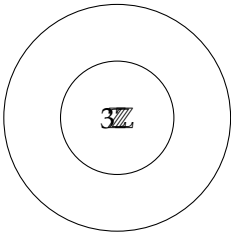
Quotient Group $\mathbb{Z}/2\mathbb{Z}$

Cosets: $\{0, 1\}$ modulo 2

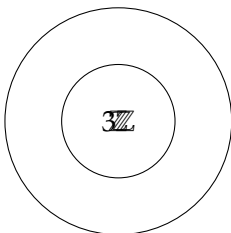


Quotient Ring $\mathbb{Z}/2\mathbb{Z}$

Elements: $\{0, 1\}$ modulo 2



Cyclic Subgroup $\langle 3 \rangle = 3\mathbb{Z}$



Principal Ideal $(3) = 3\mathbb{Z}$